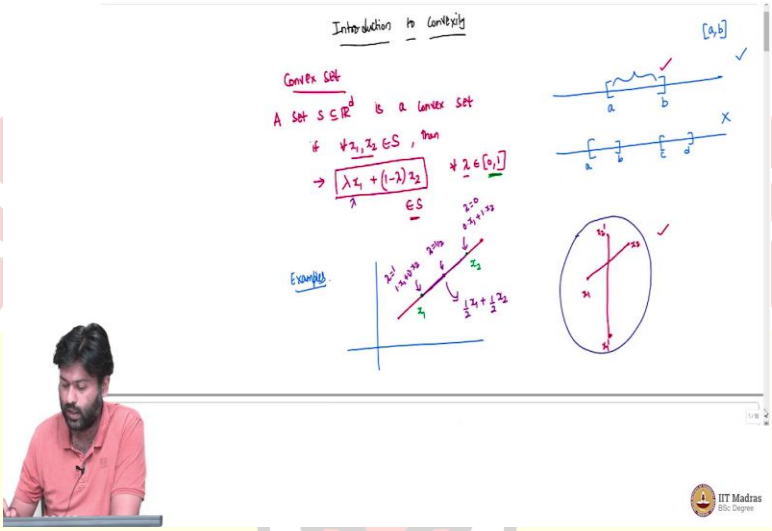


IIT Madras

ONLINE DEGREE

Professor Arun Rajmukumar
Department of Computer Science Engineering
Indian Institute of Technology, Madras
Introduction to Convexity

(Refer Slide Time: 00:31)



Welcome back everyone, today we are going to look at a very important idea, which is useful for optimization in general and specifically for optimization problems that arise in a context of machine learning and data science. So this would be the, this topic would be a brief introduction to convexity.

So, if you remember from previous lectures, what we were discussing was, we were saying how to minimize optimization functions and we said that, for instance, if you run your gradient descent algorithm, the solution that you would get would end up being a local minima or a local maxima depending on whether you are doing a gradient descent or a gradient ascent.

Now, we would ideally in a lot of situations want to find the global minima or the global maxima and I mentioned that this is not always possible. But it turns out that it is indeed possible for certain classes of functions and these classes of functions are also called as the convex functions. So, now, before getting into convex functions, let us start with a very brief introduction about what is convexity and why do we care about that and later on, we will slowly build up to the notion of convex functions and why they are interesting.

So let us begin, so what this is a brief introduction to convexity. So, what we want to do is, we want to at a high level, if you are in the real line, and let us say you look at an interval, this is between a and b and we somehow want to say that this set between a and b which we can call us a , b is a convex set whereas a set something like this, which is a b, c d. So, it is not a convex set.

So, we do not want this we want this. In this case, it looks like there is some break in between the 2 pieces of the set and, and so, we do not somehow want this. Now, the question is how can we formalize this in a more precise mathematical way. For that we first have to precisely say what do we mean by a convex set to formalize the notion that we just put down here now we are going to define what is meant by a convex set. So let me give the definition here of a convex set.

We are going to think of a set S this could be in general in Rd not necessarily in real numbers, this set was in real numbers, but we could in general think of a set in Rd and we will say is a convex set if something happens, what should happen if you take any two points in the set x1 and x2, then consider the following new point that I generate from these two points which is λ times x1 + 1 - λ times x2 I can pick any λ which is which lies between 0 and 1.

Then I want this new point that I am getting from my two points that I initially picked to also lie in the set. So, you if we can take any two points and any λ between 0 and 1 and then if you compute λ x1 + 1 - λ x2, if it is also in the set S, then we will call that set x a convex set. Now what do I even mean by this. So what does it mean to compute λ x1 + 1 - λ x2 where λ is between 0 and 1.

To see this, let us look at some examples. Examples let us say we are in two dimension and we have a set like this. So this is just a set of all points that are on this green line. Now, I take a point on this green line, which is a sorry red line, I take a point on this red line, a green point, which will call x1. Let us look at another point x2.

Now where is λ x1 + 1 - λ x2 let us say for λ is half, λ can be anything between 0 and 1. But then if you put λ equal to half, where is that point? Well, that point would exactly be in the middle of this line segment that joins x1 and x2. So this is going to be half x1 + half x2. Now, what happens when you have λ equals 0, when you have λ equals 0 this definition of λ x1 + 1 - λ x2 simply becomes x2.

So, this is simply 0 times x1 + 1 times x2, and whereas this is 1 times x1 + 0 times x2. Now, this is the case where λ is 1, this is the case where λ is 0 and this is the case where λ is half. Now, as

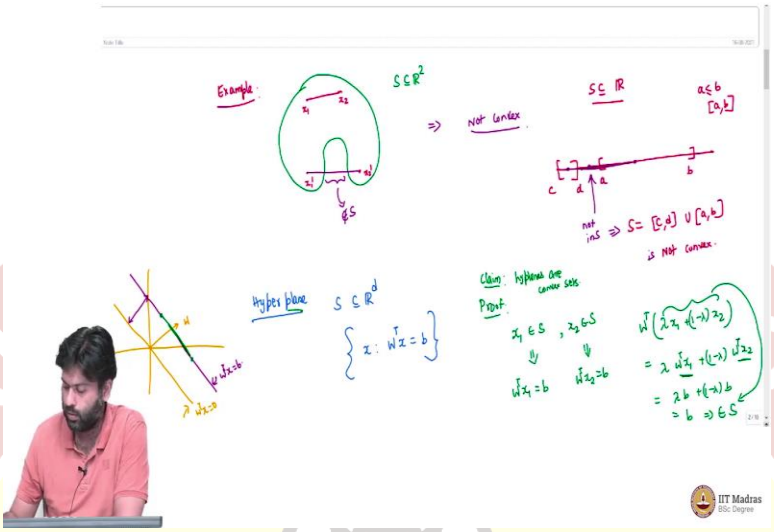
you can see, as you move λ from 1 to 0, it is you can think of it as a lever, which kind of moves this point from x1 to x2.

So, which means that if you want to look at the set of all points, which can be produced as λ x1 + 1 - λ x2, then that is exactly the set of points which lies on the line segment joining x1 and x2 and because of the way we have defined the set, and we have defined the set, which is simply the set of all red points to begin with, this line segment completely is contained in the red line bigger line segment and so this is a convex set.

So, you can do this for not just for x1 and x2 that I picked, you can do this for any two points on this red line and it is easy to convince that yourself that the line segment joining those two points should also be in the set, this simple example of a convex set. Here is another example let us look at a set like this. So think of this as a slightly tilted circle and ellipse so to say.

Now, again here if we take any two points x1 and x2 and then look at the line segment, we are always looking at the line segment that joins two points. Now, that lie that lies entirely within the set. So I can do this for any two points, the line segment will still maybe this is x1 dash x2 dash, the line segment will still lie on this inside this set. So this is also a convex set.

(Refer Slide Time: 07:25)



So, now is there an example that you can think of which is not a convex set and should not be too hard to think of an example. Let us, look at this example. If you want to try it out yourself, please pause and think about it. But here is an example of a set which is not convex. So a set something like this. This is a set S, which is a subset of R2, but why is this not convex? If I pick two points x1 and x2. Now, if we look at the line segment joining them in this particular case, it turns out that the line segment entirely is inside the set but they should happen for every two points x1 and x2 that I can pick from the set.

Now, can you think of two points in the set where if I draw the line segment it will not be in the set that is not too hard to see. So if we take x1 dash x2 dash as two points here, now if I draw the line segment joining these, well this is not entirely contained in the set S simply because this is not an S and so, this is not convex.

So, this is not a convex set of course in one dimension. Now if you had to come up with all let us say all possible convex sets in one dimension, so let us say S is a subset of R. Now, how are the convex sets going to be, the only way you can have a convex set which is a subset of R is if it is some interval a , b of course a can be even - infinity, b can even be infinity.

So, as long as a is less than b, you have less than or equal to b and you have an interval of the form a , b then this would be a convex set which means that if I take two intervals which is let us say c and d union a and b let us say this is the interval c , d union the interval a , b this is not convex.

Why is this not convex? Because simply I can take two points, one in this interval, one in this interval and then if we look at the line segment joining them, there is at least one point in this case many more infinite number of points which are not in the set. So this is not an S which implies is not convex. So this is the general idea of convex sets, what are some interesting convex sets which will be relevant to machine learning applications here is one set which we have seen several times but may not have realized that it is a convex set, but look at the set.

So, the set of the sets that we typically have been calling as half spaces. So, how do you define this? Think of this as if this is a set in Rd think of this as the set of all x such that there is some W such that W transpose x equal to let us say, maybe not even half space, let us look at the first case of a hyperplane.

So, this is the hyperplane W transpose x equals b. Now you are given some W and then you are looking at the set of all x which satisfies W transpose x equal's b. So is this a convex set? I claim it is a convex set and that needs a proof. The proof that this is convex. So the claim hyperplanes are convex sets and the proof is not too hard, it comes directly from the definition. Pick any two points x1 is in S, x2 is an S.

Now, because of the fact that the x1 in S, we know that W transpose x1 has to be b and because x2 is also an x, we know W transpose x2 also has to be a b. Now what do we have to show if I take some λ x1 + 1 - λ x2 for some λ and now basically, I am looking at the point which is on the line segment joining x1 and x2.

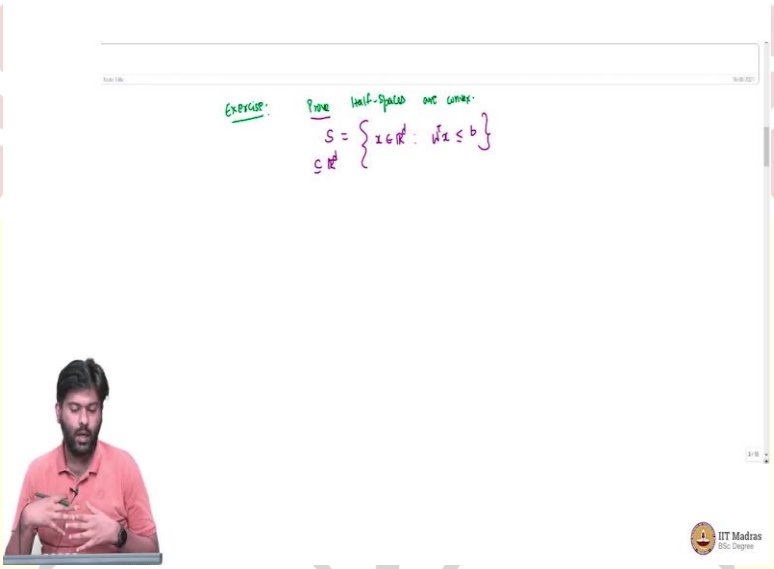
Now, what happens to W transpose this point new point this is simply λ times W transpose x1 + 1 - λ times W transpose x2. But I know already that because of the fact that x1 and x2 are on this line, this is also be the W transpose x1 is also b W transpose x2 is also b. Which means this is simply λ b + 1 - λ b, which is b.

So, which implies lies, this belongs to S which belongs to S the point λ x1 + 1 - λ x2 belongs to S. So now this satisfies the definition of convexity and so this is a convex set. So geometrically, you may already understand how this looks like. If you R2 we have some W like this and if you are also given b. We know that this is the set of all points such that W transpose x equals 0.

Now let us say b is some positive, it can be negative also. But if is some positive constant, then you are essentially pushing this line parallelly by some units, in this case, b units. So this would

be W transpose x equals b. So the geometric way of saying this is because this is like a line in R2, it necessarily has to be the case that I take any two points, the line segment will also be on this line. In more in more generally, if it is Rd there is not going to be a line, there is going to be like a plane, that way, it is called a hyperplane and you take any two points on the hyperplane that will also be on the hyper plane and so that is a convex set.

(Refer Slide Time: 13:18)



An exercise, not too hard, but I suggest you try this out. Now, prove half spaces are convex. What is a half space? It is like a hyperplane. But then it is, instead of the equality in the definition of hyperplane, we will have an inequality. So, given a set given W, we define a set S which is the set of all x in Rd which satisfies. So, S is also a subset of Rd and the definition is x the set of all x and Rd such that W transpose x is less than or equal to b.

So, remember, when this was equal to it was a hyperplane. Now we have made it less than or equal. So now, can you prove that this is a this is also a convex set. Again, the hint is use the definition and you should be able to do this. So these are some relevant sets which will keep appearing in machine learning based applications and formulations and so it would be good to understand that these are indeed convex sets.